Complex Bollobas-Erdos graphs with arbitrary rational cross density

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Editorial note. The printed bound at source964 is too strong for odd p: an open half-plane through0 contains at most ceil(p/2) p-th roots. The actual assumed clique at958 already gives |X|>=ceil((p+ell+1)/2)>ceil(p/2), because ell>=1, so the author convex-hull contradiction still follows. The printed line is retained; it is not asserted literally true for odd p.

Editorial note. In the rotation label set at962 the base vertex x0 has label0; its distance to rho^0*x0 is0. Every other vertex of X has a unique nonzero label by the root-separation lemma. This is a convention for the label set, not a loop edge added to the graph.

Editorial note. The far-pair auxiliary statement948-949 needs a clique with at least2 vertices. Its only application in the counted proof is to Y, where959 gives |Y|>=ell+1>=2. The unrestricted auxiliary statement is not counted as a separate result.

Editorial note. At939 the authors explicitly omit the upper-degree calculation. The independent reviews confirm that the already proved mu/4 cell perturbation and angle estimate925-937 apply in reverse to place neighboring cells inside a slightly enlarged phase interval, whose uniform phase measure gives the stated bound. The omission is disclosed; no generated calculation is presented as author proof.

Editorial scope. Count only GBE. All construction and local verification text is retained. Quantitative cap/partition results are explicit standard inputs; source-local rotation and mixed-clique arguments are included as full author text.

Allowed standard prerequisites (editorial).

- Spherical cap measure bounds and equal-measure small-diameter partitions. The explicit stated quantitative bounds are allowed inputs. They do not give the phase-window graph, clique exclusion or prescribed rational density.
- Spherical isoperimetry and Caratheodory theorem in the plane. Translate cap bounds to two-set separation; represent zero in a convex hull by at most three roots of unity.
- Hierarchy of asymptotic parameters. Choose K, epsilon and k in the stated order, then choose n sufficiently large; use diagonal choice to turn the displayed quantitative errors into little-o bounds.

Ramsey graphs for cliques are believed to be random-like; while on the other hand, the Turán graphs from extremal graph theory are highly structured. Initiated in 1969 by Sós, and later generalised by Erdős, Hajnal, Sós and Szemerédi [3], *Ramsey-Turán theory* combines flavours of graph Ramsey and Turán problems. The *Ramsey-Turán number* $\mathsf{RT}_p(n, K_q, m)$ is the maximum number of edges in an n-vertex K_q-free graph G with $\alpha_p(G) \leq m$, where $\alpha_p(G) = \max\{|U| : U \subseteq V(G) \text{ and } G[U] \text{ is } K_p\text{-free}\}$ is the *p-independence number* of G. Notice that when $p = 2$ and $m = n$, we recover the Turán number of K_q; and as we consider large graphs, that is $n \to \infty$, by Ramsey's theorem, m should be taken as a function of n.

The classical setting of sublinear independence number is defined as follows. Let $\varrho_p(q)$ be the *Ramsey-Turán density*:

$$\varrho_p(q) := \lim_{\varepsilon \to 0} \lim_{n \to \infty} \frac{\mathsf{RT}_p(n, K_q, \varepsilon n)}{\binom{n}{2}}.$$

The existence of the limit was shown by Erdős, Hajnal, Simonovits, Sós and Szemerédi [2]. Then $\mathsf{RT}_p(n, K_q, o(n)) := \varrho_p(q)\binom{n}{2} + o(n^2)$.

Conjecture A ([2], Conjecture 2.9)**.** *The asymptotic extremal graphs G for $\varrho_p(q)$ have the following structure. Let $q = pt + r + 2$ where $t \in \mathbb{N}$ and $0 \leq r < p$. Then there is a partition $V(G) = V_0 \cup V_1 \cup \ldots \cup V_t$ such that*

- *$e(G[V_i]) = o(n^2)$ for all $0 \leq i \leq t$;*
- *$d_G(V_0, V_1) = \frac{r+1}{p} - o(1)$, and degrees in $G[V_0, V_1]$ differ by $o(n)$;*
- *$d_G(V_i, V_j) = 1 - o(1)$ for all pairs $\{i, j\} \neq \{0, 1\}$.*

In particular,

$$\varrho_p(q) = \varrho_p^*(q) := \frac{(t-1)(2p-r-1) + r + 1}{t(2p-r-1) + r + 1}. \tag{1}$$

Notation. We write $[a, b] := \{a, \ldots, b\} \subseteq \mathbb{Z}$, for all $a, b \in \mathbb{Z}$ with $a \leq b$. Whenever $a = 1$, then we use $[b]$ instead of $[a, b]$.

We will use bold face lower case symbols, e.g. $\boldsymbol{w}, \boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z}$, for vectors in $\mathbb{C}^k$, equipped with the standard inner product $\langle \boldsymbol{w}, \boldsymbol{z} \rangle = \sum_{i \in [k]} w_i \bar{z}_i$. We write $|\boldsymbol{z}| = \sqrt{\langle \boldsymbol{z}, \boldsymbol{z} \rangle}$ for its ℓ_2-norm.

For $k \in \mathbb{N}$, let $\mathsf{S}^{k-1}(\mathbb{R}) \subseteq \mathbb{R}^k$ denote the standard $(k-1)$-dimensional real unit sphere, and write

$$\mathsf{S}^{k-1}(\mathbb{C}) = \left\{ (z_1, \ldots, z_k) \in \mathbb{C}^k : \sum_{i=1}^{k} |z_i|^2 = 1 \right\}$$

for the $(k-1)$-dimensional complex unit sphere. As the map

$$\varphi : \ (x_1 + iy_1, \ldots, x_k + iy_k) \longmapsto (x_1, y_1, x_2, y_2 \ldots, x_k, y_k) \tag{2}$$

from $\mathsf{S}^{k-1}(\mathbb{C})$ to $\mathsf{S}^{2k-1}(\mathbb{R})$ is an invertible isometry, various properties of high dimensional real spheres extend naturally to the complex ones.

Throughout the paper, when given a high dimensional unit sphere, we will write λ for the Lebesgue measure, normalised so that the unit sphere has measure 1. For two subsets of a unit sphere A and B, denote by $d_{\max}(A,B) := \sup\{|\boldsymbol{a}-\boldsymbol{b}| : \boldsymbol{a}\in A,\ \boldsymbol{b}\in B\}$ the Euclidean distance between them. In the case $A=B$, write $\mathrm{diam}(A) := d_{\max}(A,A)$ for the *diameter* of A.

A *spherical cap* is the smaller intersection of the unit sphere with a half-space. Given a spherical cap C bounded by some hyperplane H, we call the point in C with maximum Euclidean distance to H the *centre* of the spherical cap. The distance from the centre to H is the *height* of the spherical cap. Note that $\mathrm{diam}(C)$ is just the diameter of the intersection of C and H.

We will use the following lower and upper bounds on the measure of spherical caps. They follow from the known results of the real sphere and the use of the isometry φ in (2).

Lemma 0.1 ([1])**.** *For all $\delta>0$ and integers $k\geq 3$, let $B\subseteq \mathsf{S}^{k-1}(\mathbb{C})$ be the spherical cap consisting of all points with distance at most $\sqrt{2}-\delta/\sqrt{2k}$ from a fixed point in $\mathsf{S}^{k-1}(\mathbb{C})$. Then $\lambda(B)\geq 1/2-\sqrt{2}\delta$.*

Lemma 0.2 ([6])**.** *Let $\alpha\in[0,1)$ and $C\subseteq \mathsf{S}^{k-1}(\mathbb{C})$ be a spherical cap with height $1-\alpha$. Then $\lambda(C)\leq e^{-k\alpha^2}$.*

Recall that a spherical cap with height $1-\alpha$ has diameter $2\sqrt{1-\alpha^2}$. It is a simple consequence (see [1]) of the isoperimetric inequality for spheres [5], that for any sets $A,B,C\subseteq \mathsf{S}^{k-1}(\mathbb{C})$ of equal measure, if C is a spherical cap, then $d_{\max}(A,B)\geq \mathrm{diam}(C)$. Altogether, we have the following 2-set version of Lemma 0.2.

Lemma 0.3. *Let $\nu\in(0,1)$ and $A,B\subseteq \mathsf{S}^{k-1}(\mathbb{C})$ with $\lambda(A),\lambda(B)>e^{-k\nu/2}$, then $d_{\max}(A,B)\geq 2-\nu$.*

The following folklore result partitions the sphere into small pieces of equal measure (see e.g. [4]).

Lemma 0.4. *There exists $C>0$ such that the following holds. Let $0<\delta<1$ and $n\geq (C/\delta)^k$. Then $\mathsf{S}^{k-1}(\mathbb{R})$ can be partitioned into n pieces of equal measure, each of diameter at most δ.*

Theorem 0.5 (Complex Bollobás-Erdős graph)**.** *Let p,ℓ be integers with $1\leq \ell<p$. Then for all sufficiently large n, there exists a graph G with vertex partition $W\cup Z$, where $|W|=|Z|=n$, such that $\alpha_p(G)=o(n)$, $e(G[W]),e(G[Z])=o(n^2)$, and $e_G(W,Z)=(\ell/p-o(1))n^2$. If additionally $\ell\leq p/2$, then G is $K_{p+\ell+1}$-free, and consequently,*

$$\varrho_p(p+\ell+1)\geq \frac{\ell}{2p}=\varrho_p^*(p+\ell+1).$$

Fix integers $1\leq \ell<p$. For Theorem 0.5, we will construct a graph G with vertex partition $W\cup Z$ where $|W|=|Z|=n$, satisfying the following:

A1 $\alpha_p(G)=o(n)$;

A2 $e(G[W]),e(G[Z])=o(n^2)$;

A3 $e(G)=\left(\frac{\ell}{p}-o(1)\right)n^2$;

A4 if $\ell\leq p/2$, then G is $K_{p+\ell+1}$-free.

0.1 Construction

Choose constants

$$0 < 1/k \ll \varepsilon \ll 1/K \ll 1/p, \quad \mu := \varepsilon/\sqrt{2k}, \quad \text{and} \quad n \geq \left(\frac{4C_{0.4}}{\mu}\right)^{2k}, \quad (3)$$

where $C_{0.4}$ is the constant obtained from Lemma 0.4. Using the isometry φ in (2) and Lemma 0.4, we can partition $\mathsf{S}^{k-1}(\mathbb{C})$ into n domains $D_1, \ldots, D_n$ with equal measure and diameter at most $\frac{\mu}{4}$. Next, for all $i \in [n]$, choose two arbitrary points $\boldsymbol{w}_i, \boldsymbol{z}_i \in D_i$, and let $W := \{\boldsymbol{w}_1, \ldots, \boldsymbol{w}_n\}$ and $Z := \{\boldsymbol{z}_1, \ldots, \boldsymbol{z}_n\}$. Set $\rho := \cos(2\pi/p) + i\sin(2\pi/p)$ to be the primitive p-th root of unity. Note that $\boldsymbol{w} \mapsto \rho\boldsymbol{w}$ rotates $\mathsf{S}^{k-1}(\mathbb{C})$. The edge set of G is defined as follows (see also Figure 1).

B1 Two vertices $\boldsymbol{w}, \boldsymbol{w}' \in W$ form an edge if and only if there exists $h \in [p-1]$ such that

$$\left|\boldsymbol{w} - \rho^h \boldsymbol{w}'\right| \leq \sqrt{\mu}.$$

Define $E(G[Z])$ similarly. For such a pair, we say $\boldsymbol{w}$ is an *h-rotation* of $\boldsymbol{w}'$.

B2 A cross pair $(\boldsymbol{w}_i, \boldsymbol{z}_j) \in W \times Z$ forms an edge if and only if the following hold.

(i) For all $h \in \mathbb{Z}_p$,

$$\left|\mathrm{Im}(\rho^h \langle \boldsymbol{w}_i, \boldsymbol{z}_j \rangle)\right| \geq K\mu.$$

(ii) There exists $\alpha \in [0, \frac{2\pi\ell}{p}]$ such that

$$e^{-i\alpha}\langle \boldsymbol{w}_i, \boldsymbol{z}_j \rangle \in [0, 1].$$

0.2 Structure of the inner graphs

In this subsection, using isoperimetry and concentration of measure, we shall derive that the inner graphs $G[W], G[Z]$ are K_{p+1}-free graphs (Lemma 0.6) with sublinear p-independence number (Lemma 0.7) and zero edge density (Lemma 0.9), thus verifying **A1** and **A2**.

Lemma 0.6. *Let $\boldsymbol{w}_i, \boldsymbol{w}_j, \boldsymbol{w}_t \in W$ span a triangle in G. If $\boldsymbol{w}_i, \boldsymbol{w}_j$ are an h_i- and h_j-rotation of $\boldsymbol{w}_t$ respectively, then $\boldsymbol{w}_i$ is an $(h_i - h_j)$-rotation of $\boldsymbol{w}_j$ and $h_i \neq h_j$.*

Consequently, $G[W]$ is K_{p+1}-free. The same holds for Z.

Proof. For any $m \in [p-1]$, we have

$$|1 - \rho^m|^2 = 2 - 2\cos(2\pi m/p) \geq 2 - 2\cos(2\pi/p) = 4\sin^2(\pi/p) \geq (4/p)^2, \quad (4)$$

as $\sin x \geq \frac{2x}{\pi}$ for all $x \in [0, \frac{\pi}{2}]$ by concavity. By **B1**, there is $h \in [p-1]$ such that $\boldsymbol{w}_i$ is an h-rotation of $\boldsymbol{w}_j$. Recall that vertices of G, viewed as points in $\mathsf{S}^{k-1}(\mathbb{C})$, all have modulus 1. So

$$\begin{aligned}|1 - \rho^{h_i - h_j - h}| &= |\rho^h \boldsymbol{w}_j - \rho^{h_i - h_j}\boldsymbol{w}_j| \leq |\boldsymbol{w}_i - \rho^h \boldsymbol{w}_j| + |\boldsymbol{w}_i - \rho^{h_i}\boldsymbol{w}_t| + |\rho^{h_i}\boldsymbol{w}_t - \rho^{h_i - h_j}\boldsymbol{w}_j| \\ &= |\boldsymbol{w}_i - \rho^h \boldsymbol{w}_j| + |\boldsymbol{w}_i - \rho^{h_i}\boldsymbol{w}_t| + |\boldsymbol{w}_j - \rho^{h_j}\boldsymbol{w}_t| \leq 3\sqrt{\mu},\end{aligned}$$

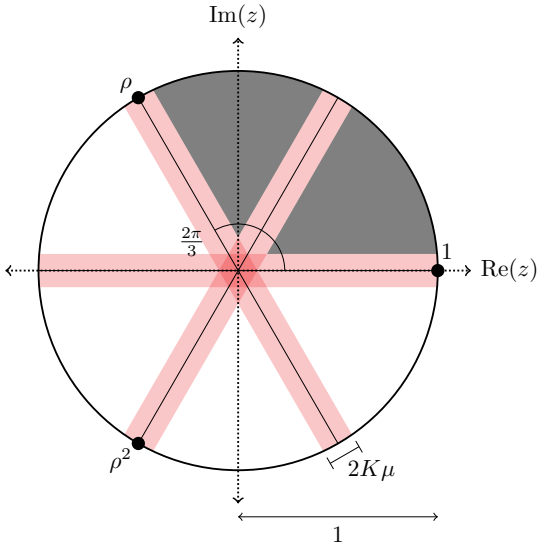


Figure 1: An illustration of the position of $\langle \boldsymbol{w}_i, \boldsymbol{z}_j \rangle$ for $p = 3$. The pink stripes are the ones excluded in **B2**(i); while the dark regions correspond to **B2**(ii).

which together with (3) and (4) implies $h = h_i - h_j$. Since $h \neq 0$, we have that $h_i \neq h_j$.

Suppose that $\boldsymbol{w}_0, \boldsymbol{w}_1, \ldots, \boldsymbol{w}_p$ span a clique in $G[W]$. Then by **B1** there are $h_1, \ldots, h_p \in [p-1]$ such that $\boldsymbol{w}_i$ is an h_i-rotation of $\boldsymbol{w}_0$ for all $i \in [p]$. By the Pigeonhole Principle, there are distinct $i, j \in [p]$ such that $h_i = h_j$, contradicting the first part. □

Lemma 0.7. *Every set $X \subseteq W$ with $|X| \geq pe^{-\mu k/40} \cdot |W|$ contains a copy of K_p. In particular,*

$$\alpha_p(G) \leq 2pe^{-\mu k/40} n.$$

For its proof, we need the following consequence of concentration of measure.

Lemma 0.8. *Let $p \geq 2$ be an integer, $\nu \leq \min\{\frac{16}{p^2}, 1\}$ and $A \subseteq \mathsf{S}^{k-1}(\mathbb{C})$ with $\lambda(A) \geq pe^{-\nu k/32}$. Then there are distinct points $\boldsymbol{a}_0, \ldots, \boldsymbol{a}_{p-1} \in A$ such that, for all $h, m \in \mathbb{Z}_p$,*

$$\left|\rho^h \boldsymbol{a}_h - \rho^m \boldsymbol{a}_m\right| < \sqrt{\nu}.$$

Proof. For all $h \in [p-1]$, define

$$A_h := \left\{\boldsymbol{a} \in A : \ \forall \boldsymbol{a}' \in A, \ \left|\boldsymbol{a} + \rho^h \boldsymbol{a}'\right| < 2 - \nu/16\right\}.$$

Note that $\lambda(A_h) \leq e^{-\nu k/32}$ as otherwise, the sets A_h and $-\rho^h A$ violate Lemma 0.3. Therefore, $A \neq A_1 \cup \cdots \cup A_{p-1}$, and we can pick a point $\boldsymbol{a}_0 \in A \setminus (A_1 \cup \cdots \cup A_{p-1})$.

For all $h \in [p-1]$, since $\boldsymbol{a}_0 \notin A_h$, there exists a point $\boldsymbol{a}_h \in A$ with $|\boldsymbol{a}_0 + \rho^h \boldsymbol{a}_h| \geq 2 - \nu/16$. We claim that $|\boldsymbol{a}_0 - \rho^h \boldsymbol{a}_h| < \sqrt{\nu}/2$, for all $h \in \mathbb{Z}_p$. The inequality is trivial for $h = 0$. For $h \in [p-1]$, it follows from the parallelogram law:

$$|\boldsymbol{a}_0 - \rho^h \boldsymbol{a}_h|^2 = 2(|\boldsymbol{a}_0|^2 + |\rho^h \boldsymbol{a}_h|^2) - |\boldsymbol{a}_0 + \rho^h \boldsymbol{a}_h|^2 \leq 4 - (2 - \nu/16)^2 < \nu/4.$$

Thus, for all $h, m \in \mathbb{Z}_p$, by the triangle inequality, we obtain

$$|\rho^h \boldsymbol{a}_h - \rho^m \boldsymbol{a}_m| \leq |\rho^h \boldsymbol{a}_h - \boldsymbol{a}_0| + |\boldsymbol{a}_0 - \rho^m \boldsymbol{a}_m| < \sqrt{\nu}. \tag{5}$$

We are left to show that all points $\boldsymbol{a}_0, \ldots, \boldsymbol{a}_{p-1}$ are distinct. Suppose to the contrary that for some distinct $h, m \in \mathbb{Z}_p$, $\boldsymbol{a}_h = \boldsymbol{a}_m$. Then, as in (4),

$$|\rho^h \boldsymbol{a}_h - \rho^m \boldsymbol{a}_m| = |\rho^h - \rho^m| = |\rho^{h-m} - 1| \geq 4/p \geq \sqrt{\nu},$$

a contradiction to (5). □

We are now ready to prove Lemma 0.7.

Proof of Lemma 0.7. Let $X \subseteq W$ with $|X| \geq pe^{-\mu k/40} \cdot |W|$ and let $A := \bigcup\{D_i : \boldsymbol{w}_i \in X\}$; then $\lambda(A) = |X|/|W| \geq pe^{-\mu k/40}$. By Lemma 0.8, there exist distinct $\boldsymbol{a}_0, \ldots, \boldsymbol{a}_{p-1} \in A$ such that $|\rho^h \boldsymbol{a}_h - \rho^m \boldsymbol{a}_m| \leq \sqrt{4\mu/5}$ for all $h, m \in \mathbb{Z}_p$. For each $h \in \mathbb{Z}_p$, let $\boldsymbol{x}_h \in X$ be the vertex lying in the same domain D_i as $\boldsymbol{a}_h$, i.e. $|\boldsymbol{x}_h - \boldsymbol{a}_h| \leq \mu/4$.

Now, for distinct integers $h, m \in \mathbb{Z}_p$, by triangle inequality and (3), we get

$$|\boldsymbol{x}_h - \rho^{m-h}\boldsymbol{x}_m| = |\rho^h \boldsymbol{x}_h - \rho^m \boldsymbol{x}_m| \leq |\rho^h \boldsymbol{x}_h - \rho^h \boldsymbol{a}_h| + |\rho^h \boldsymbol{a}_h - \rho^m \boldsymbol{a}_m| + |\rho^m \boldsymbol{a}_m - \rho^m \boldsymbol{x}_m| \leq \sqrt{\mu}.$$

Thus $\boldsymbol{x}_h \boldsymbol{x}_m \in E(G[W])$ and $\boldsymbol{x}_0, \ldots, \boldsymbol{x}_{p-1}$ induce a copy of K_p, finishing the proof. □

The fact that the inner graphs $G[W], G[Z]$ have zero edge density follows already from their being K_{p+1}-free and having sublinear p-independence number, as then their maximum degree is at most $\alpha_p(G[W]) = e^{-\Theta(\sqrt{k})}n$ by Lemma 0.7. We can in fact give a tighter bound via a direct estimation.

Lemma 0.9. *The maximum degree of $G[W], G[Z]$ is at most $pe^{-k(1-\mu)^2}n \leq e^{-k/2}n$.*

Proof. By the construction of G, in particular **B1** and that each domain D_i has diameter at most $\mu/4$, we see that in $G[W]$ every vertex has degree $n(p-1)$ times the measure of a spherical cap whose points are within distance $d = \sqrt{\mu} \pm \mu/4$ from its centre. As the height of such a cap is precisely $d^2/2$, the conclusion follows from Lemma 0.2. □

0.3 Angle decides cross density

In this subsection, we verify **A3**.

Lemma 0.10. *Every vertex in W has $(\frac{\ell}{p} \pm \frac{1}{\sqrt{K}})n$ neighbours in Z, and vice versa.*

Proof. For each point $\boldsymbol{x} \in \mathsf{S}^{k-1}(\mathbb{C})$, define sets

$$J(\boldsymbol{x}) := \left\{\boldsymbol{y} \in \mathsf{S}^{k-1}(\mathbb{C}) : \left|\mathrm{Im}\langle \boldsymbol{x}, \rho^h \boldsymbol{y}\rangle\right| > \left(K + \tfrac{1}{4}\right)\mu, \text{ for all } h \in \mathbb{Z}_p\right\}, \text{ and}$$

$$I(\boldsymbol{x}) := \left\{\boldsymbol{y} \in J(\boldsymbol{x}) : \arg\langle \boldsymbol{x}, \boldsymbol{y}\rangle \in \left[1/\sqrt{K},\ 2\pi\ell/p - 1/\sqrt{K}\right]\right\}.$$

We shall show that for each vertex $\boldsymbol{w} \in W$, vertices whose domains intersect the associated set $I(\boldsymbol{w})$ are adjacent to $\boldsymbol{w}$. That is,

$$Z_{\boldsymbol{w}} := \{\boldsymbol{z}_j \in Z : D_j \cap I(\boldsymbol{w}) \neq \varnothing\} \subseteq N_G(\boldsymbol{w}). \tag{6}$$

We first bound the measure of this associated set.

Claim 0.11. *For any $\boldsymbol{x} \in \mathsf{S}^{k-1}(\mathbb{C})$, we have $\lambda(I(\boldsymbol{x})) \geq \frac{\ell}{p} - \frac{1}{\sqrt{K}}$.*

Proof of claim. Fix an arbitrary $\boldsymbol{x} \in \mathsf{S}^{k-1}(\mathbb{C})$. Define

$$L := \left\{ \boldsymbol{y} \in \mathsf{S}^{k-1}(\mathbb{C}) : |-\boldsymbol{x}i - \boldsymbol{y}| \leq \sqrt{2} - K\mu \right\}.$$

By Lemma 0.1 (with $\delta = \varepsilon K$), we see that $\lambda(L) \geq \frac{1}{2} - \sqrt{2}\varepsilon K$. For each $y \in L$, using (3),

$$2 - 2\operatorname{Im}\langle \boldsymbol{x}, \boldsymbol{y} \rangle = 2 - 2\operatorname{Re}\langle -\boldsymbol{x}i, \boldsymbol{y} \rangle = |-\boldsymbol{x}i - \boldsymbol{y}|^2 \leq (\sqrt{2} - K\mu)^2 < 2 - 2\left(K + \tfrac{1}{4}\right)\mu,$$

and so $\operatorname{Im}\langle \boldsymbol{x}, \boldsymbol{y} \rangle > (K + \frac{1}{4})\mu$. Thus, by symmetry, the measure of the set

$$\left\{ \boldsymbol{y} \in \mathsf{S}^{k-1}(\mathbb{C}) : \left|\operatorname{Im}\langle \boldsymbol{x}, \boldsymbol{y} \rangle\right| \leq \left(K + \tfrac{1}{4}\right)\mu \right\}$$

is at most $1 - 2\lambda(L) \leq 1 - 2\left(\frac{1}{2} - \sqrt{2}\varepsilon K\right) = 2\sqrt{2}\varepsilon K$. Since $\langle \boldsymbol{x}, \rho^h \boldsymbol{y} \rangle = \langle \rho^{-h}\boldsymbol{x}, \boldsymbol{y} \rangle$, we can take the union bound of such sets over $\boldsymbol{x}$, $\rho\boldsymbol{x}$,..., $\rho^{p-1}\boldsymbol{x}$ to deduce that $\lambda(\overline{J(\boldsymbol{x})}) \leq 2\sqrt{2}\varepsilon K p$, where $\overline{J(\boldsymbol{x})} := \mathsf{S}^{k-1}(\mathbb{C}) \setminus J(\boldsymbol{x})$. Therefore, by (3), we get

$$\lambda(I(\boldsymbol{x})) \geq \frac{1}{2\pi}\left(\frac{2\pi\ell}{p} - \frac{2}{\sqrt{K}}\right) - \lambda(\overline{J(\boldsymbol{x})}) \geq \frac{\ell}{p} - \frac{1}{\sqrt{K}}.$$

■

Fix a vertex $\boldsymbol{w} \in W$ and let $Z_{\boldsymbol{w}}$ be as in (6). As each domain D_j has measure $\frac{1}{n}$, the above claim entails

$$|Z_{\boldsymbol{w}}| \geq n \sum_{\boldsymbol{z}_j \in Z_{\boldsymbol{w}}} \lambda(D_j \cap I(\boldsymbol{w})) = n \cdot \lambda(I(\boldsymbol{w})) \geq \left(\frac{\ell}{p} - \frac{1}{\sqrt{K}}\right) n.$$

We are left to show $Z_{\boldsymbol{w}} \subseteq N_G(\boldsymbol{w})$ and that the upper bound on the degrees can be obtained similarly.

Fix a vertex $\boldsymbol{z}_j \in Z_{\boldsymbol{w}}$ and take a point $\boldsymbol{z}^* \in D_j \cap I(\boldsymbol{w})$. As $\boldsymbol{z}_j, \boldsymbol{z}^* \in D_j$, $|\boldsymbol{z}^* - \boldsymbol{z}_j| \leq \mu/4$. For any $h \in \mathbb{Z}_p$, using the triangle inequality, that $\boldsymbol{z}^* \in I(\boldsymbol{w})$, and the Cauchy–Schwarz inequality, we see that

$$\begin{aligned}
|\operatorname{Im}(\rho^h \langle \boldsymbol{w}, \boldsymbol{z}_j \rangle)| = |\operatorname{Im}\langle \boldsymbol{w}, \rho^{-h}\boldsymbol{z}_j \rangle| &\geq |\operatorname{Im}\langle \boldsymbol{w}, \rho^{-h}\boldsymbol{z}^* \rangle| - |\operatorname{Im}\langle \boldsymbol{w}, \rho^{-h}\boldsymbol{z}^* \rangle - \operatorname{Im}\langle \boldsymbol{w}, \rho^{-h}\boldsymbol{z}_j \rangle| \\
&\geq (K + \tfrac{1}{4})\mu - |\langle \boldsymbol{w}, \rho^{-h}\boldsymbol{z}^* \rangle - \langle \boldsymbol{w}, \rho^{-h}\boldsymbol{z}_j \rangle| \\
&\geq (K + \tfrac{1}{4})\mu - |\boldsymbol{w}| \cdot |\rho^{-h}\boldsymbol{z}^* - \rho^{-h}\boldsymbol{z}_j| \\
&= (K + \tfrac{1}{4})\mu - |\boldsymbol{z}^* - \boldsymbol{z}_j| \geq K\mu,
\end{aligned}$$

verifying **B2**(i).

For **B2**(ii), let $x := \langle \boldsymbol{w}, \boldsymbol{z}_j \rangle$ and $y := \langle \boldsymbol{w}, \boldsymbol{z}^* \rangle$. Then as above we have $|y| \geq |\operatorname{Im}\langle \boldsymbol{w}, \boldsymbol{z}^* \rangle| \geq (K + \frac{1}{4})\mu$, $|x - y| \leq |\boldsymbol{w}||\boldsymbol{z}^* - \boldsymbol{z}_j| \leq \mu/4$ and so $|x| \geq |y| - |x - y| \geq K\mu$. Consider the triangle in the complex plane with vertices $0, x, y$ and let α be its angle at 0. Then the law of cosines implies that

$$\cos\alpha = \frac{|x|^2 + |y|^2 - |x - y|^2}{2|x||y|} \geq \frac{(1 - \frac{1}{8K})(|x|^2 + |y|^2)}{2|x||y|} \geq 1 - \frac{1}{8K}.$$

As $\cos\theta \leq 1 - \frac{\theta^2}{4}$ when $\theta \in (0, 1)$, we get $|\arg\langle \boldsymbol{w}, \boldsymbol{z}_j \rangle - \arg\langle \boldsymbol{w}, \boldsymbol{z}^* \rangle| = \alpha \leq 1/\sqrt{2K}$. Therefore $\arg\langle \boldsymbol{w}, \boldsymbol{z}_j \rangle \in [0, 2\pi\ell/p]$ as $\boldsymbol{z}^* \in I(\boldsymbol{w})$, implying **B2**(ii).

We omit the proof of the upper bound on degrees since it is very similar but easier, noting that the upper bound corresponding to Claim 0.11 is clear as the set of $\boldsymbol{y} \in \mathsf{S}^{k-1}(\mathbb{C})$ with $\arg\langle \boldsymbol{x}, \boldsymbol{y} \rangle \in [s, t]$ trivially has measure at most $\frac{t-s}{2\pi}$. □

0.4 Clique number of G

Finally, we verify **A4**. We will need the following lemma.

Lemma 0.12. *Let U be a subset of W or Z such that $G[U]$ is a clique. Then there exists a pair of vertices $\boldsymbol{u}, \boldsymbol{u}' \in U$ such that $\boldsymbol{u}$ is an h-rotation of $\boldsymbol{u}'$ with $\min\{|U|-1, \lfloor p/2 \rfloor\} \le h \le \lfloor p/2 \rfloor$.*

Proof. Note that the upper bound is trivial, as in every pair of adjacent vertices, taking the smaller angle we see that one is an h-rotation of the other for some $h \le \lfloor p/2 \rfloor$. For the lower bound, set $u := |U|$. We will prove the case when $u - 1 \le \lfloor p/2 \rfloor$. The other case can be reduced to this case by taking a subset of U of size $\lfloor p/2 \rfloor + 1$.

Fix a vertex $\boldsymbol{u}_0 \in U$. By **B1**, every vertex $\boldsymbol{u} \in U \setminus \{\boldsymbol{u}_0\}$ is an h-rotation of $\boldsymbol{u}_0$, for some $h \in [p-1]$. We may assume that $h \in \{-u+2, \dots, u-2\} \setminus \{0\}$, for otherwise, $\boldsymbol{u}_0, \boldsymbol{u}$ is the pair we seek. Pair up all elements in $\{-u+2, \dots, u-2\} \setminus \{0\}$ such that their difference is $u-1$, that is, partition it into pairs $\{-u+1+j, j\}$, for all $j \in [u-2]$. We say a vertex in $U \setminus \{\boldsymbol{u}_0\}$ *belongs to the j-th pair* if it is either a $(-u+1+j)$-rotation or a j-rotation of $\boldsymbol{u}_0$. By the Pigeonhole Principle, there exists a pair of vertices $\boldsymbol{u}, \boldsymbol{u}' \in U \setminus \{\boldsymbol{u}_0\}$ that form the j-th pair for some $j \in [u-2]$. By Lemma 0.6, they are not the same rotation of $\boldsymbol{u}_0$. Therefore, without loss of generality, we can assume that $\boldsymbol{u}$ is a j-rotation of $\boldsymbol{u}_0$ and $\boldsymbol{u}'$ is a $(-u+1+j)$-rotation of $\boldsymbol{u}_0$. Thus, by Lemma 0.6, $\boldsymbol{u}$ is a $(u-1)$-rotation of $\boldsymbol{u}'$. □

Take $\ell \le p/2$. Suppose to the contrary that G contains a copy of $K_{p+\ell+1}$ on vertex set $X \cup Y$, with, say, $X \subseteq W$, $Y \subseteq Z$ and $|X| \ge |Y|$. So $\frac{p}{2} < |X| \le p$ (since X is K_{p+1}-free) and hence $|Y| \ge \ell + 1$. Let $\boldsymbol{x}_0 \in X$ be arbitrary, set

$$A := \{h \in \mathbb{Z}_p : X \text{ contains an } h\text{-rotation of } \boldsymbol{x}_0\} \quad \text{and} \quad B := \{\rho^h : h \in A\} \subseteq \{z \in \mathbb{C} : |z| = 1\}.$$

If the closed convex hull of B fails to contain 0, then B is contained in an open half-plane of $\mathbb{C}$ bounded by a line passing through 0, so we have $|X| = |B| \le \frac{p}{2}$, a contradiction. Thus 0 lies in the closed convex hull of B. Consequently, by Carathéodory's theorem, there are $h_1, h_2, h_3 \in A$ such that 0 is in the closed triangle with vertices $\rho^{h_1}, \rho^{h_2}, \rho^{h_3}$. In other words, there are reals $\lambda_1, \lambda_2, \lambda_3 \in [0,1]$ such that $\lambda_1 + \lambda_2 + \lambda_3 = 1$ and $\lambda_1\rho^{h_1} + \lambda_2\rho^{h_2} + \lambda_3\rho^{h_3} = 0$. Pick $\boldsymbol{x}_1, \boldsymbol{x}_2, \boldsymbol{x}_3 \in X$ such that $|\boldsymbol{x}_i - \rho^{h_i}\boldsymbol{x}_0| \le \sqrt{\mu}$ for all $i \in [3]$. By the triangle inequality

$$\boldsymbol{x} := \lambda_1\boldsymbol{x}_1 + \lambda_2\boldsymbol{x}_2 + \lambda_3\boldsymbol{x}_3 = \lambda_1(\boldsymbol{x}_1 - \rho^{h_1}\boldsymbol{x}_0) + \lambda_2(\boldsymbol{x}_2 - \rho^{h_2}\boldsymbol{x}_0) + \lambda_3(\boldsymbol{x}_3 - \rho^{h_3}\boldsymbol{x}_0)$$

satisfies $|\boldsymbol{x}| \le \sqrt{\mu}$. By Lemma 0.12, there exist a pair of vertices $\boldsymbol{y}, \boldsymbol{y}' \in Y$ and an integer m such that $\ell \le m \le \frac{p}{2}$ and $|\boldsymbol{y} - \rho^m\boldsymbol{y}'| \le \sqrt{\mu}$. Now,

$$\mathrm{Im}\langle \boldsymbol{x}, \boldsymbol{y} - \rho^m\boldsymbol{y}'\rangle \le |\langle \boldsymbol{x}, \boldsymbol{y} - \rho^m\boldsymbol{y}'\rangle| \le |\boldsymbol{x}||\boldsymbol{y} - \rho^m\boldsymbol{y}'| \le \mu. \tag{7}$$

On the other hand, **B2**(ii) implies that $0 \le \arg\langle \boldsymbol{x}_i, \boldsymbol{y}\rangle, \arg\langle \boldsymbol{x}_i, \boldsymbol{y}'\rangle \le \frac{2\pi\ell}{p}$. So

$$0 \le \arg(-\rho^{-m}\langle \boldsymbol{x}_i, \boldsymbol{y}'\rangle) \le 2\pi\left(\frac{\ell}{p} + \frac{p/2 - m}{p}\right) \le \pi$$

since $\ell \le m$. So the imaginary parts of $\langle \boldsymbol{x}_i, \boldsymbol{y}\rangle, -\rho^{-m}\langle \boldsymbol{x}_i, \boldsymbol{y}'\rangle$ are positive, and **B2**(i) yields

$$\mathrm{Im}\langle \boldsymbol{x}_i, \boldsymbol{y}\rangle, \mathrm{Im}(-\rho^{-m}\langle \boldsymbol{x}_i, \boldsymbol{y}'\rangle) \ge K\mu \quad \text{for all } i \in [3],$$

whence $\mathrm{Im}\langle \boldsymbol{x}, \boldsymbol{y} - \rho^m \boldsymbol{y}' \rangle \geq 2K\mu$. This contradiction to (7) concludes the proof of **A4**.

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